

OpenCPU

GCC Installation Guide

GSM/GPRS Module Series

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About the Document

History

Revision	Date	Author	Description
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1 Introduction

This document mainly introduces how to establish GCC compiler environment for Windows, and how to compile App in OpenCPU SDK using GCC.

Quectel
Preliminary

2 Installation

OpenCPU uses “Sourcery CodeBench Lite” as GCC compiler. Developers can freely download the latest version from the [Sourcery CodeBench web site](#).

2.1. System Requirements

This version of Sourcery CodeBench supports the following host operating systems and architectures:

- Microsoft Windows XP (SP1 or later)
- Windows Vista
- Windows 7 systems using IA32, AMD64, and Intel 64 processors

In order to install and use Sourcery CodeBench Lite, you must have at least 512MB of available memory.

2.2. Download GCC Installer

Step 1: Create an account in the <http://www.mentor.com/profile/sign-up>.

Step 2: Open the download website “http://www.codesourcery.com/gnu_toolchains/”, and select the “**Download the EABI Release for ARM Processors**”.

ARM Processors

These releases support ARM, Thumb, and Thumb-2 compilations for all architectures in active use, including version 7 of the ARM architecture.

[Read the FAQ →](#)

[Download the GNU/Linux Release →](#)

[Download the EABI Release →](#)

[Download the SymbianOS release →](#)

Figure 1: GCC Download Link

Step 3: An account is required to log in the Figure 2, and then click “**Get Lite!**”.

Sourcery CodeBench Lite Edition for ARM EABI

In partnership with leading manufacturers, Sourcery CodeBench Lite Edition delivers command-line only tools, including:

- GNU C and C++ compilers
- GNU assembler and linker
- C and C++ runtime libraries
- GNU debugger

First Name

Last Name

Email

Country

A valid email address is required.

Please Provide Your City

Get Lite!

Figure 2: Sign In

Step 4: After clicking “**Get Lite!**”, a mail from Mentor will be sent out. After you receive the email, open the mail and click the link in email, then you can get the version list.

Thank you for registering to download **Sourcery CodeBench Lite Edition for ARM EABI**.

Download your Lite edition copy of Sourcery CodeBench now:
<http://go.mentor.com/36g95>

Figure 3: Mail from Mentor

Step 5: Select the latest lite version and download it. In next step, you will face the choices below.

Sourcery CodeBench Lite Edition for ARM EABI hosted on IA32 Windows, IA32 GNU/Linux

Recommended Release

This is a fully-validated release.

[Download Sourcery CodeBench Lite 2012.09-63](#)

Available Releases

This table lists all releases for download.

Release	Status	Date
Sourcery CodeBench Lite 2012.09-63	Release	2012-11-13
Sourcery CodeBench Lite 2012.03-56	Release	2012-06-11
Sourcery CodeBench Lite 2011.09-69	Release	2011-12-19
Sourcery G++ Lite 2011.03-42	Release	2011-05-02
Sourcery G++ Lite 2010.09-51	Release	2010-11-10
Sourcery G++ Lite 2010q1-188	Release	2010-04-23
Sourcery G++ Lite 2009q3-68	Release	2009-10-20

Figure 4: GCC Version List

Step 6: Click “*IA32 Windows Installer*” to download.

Software

Download	MD5 Checksum
Recommended Packages	
IA32 GNU/Linux Installer	25c5dce1fd93410cde1c76f282ca1ed0
IA32 Windows Installer	a01ef7d89fb1a4301f1d5dfcf5318f2d
Advanced Packages	
IA32 GNU/Linux TAR	9d206de1c74f9454e468ddcdd72c9c53
IA32 Windows TAR	10fe6bc1cedda2ad04f5019e6784a88a
Source TAR	4c791ddbb3d4bcfee29a26eb4db5a244

Figure 5: Download IA32 Windows Installer

After downloading it successfully, you can get the setup package shown below.



Figure 6: Setup Package

2.3. Install GCC

Step 1: Running the Installer - Double click the GCC installer to start to install GCC. After the installer starts, follow the on-screen dialogs to install Sourcery CodeBench Lite. The installer is intended to be self-explanatory and on most pages the defaults are appropriate.

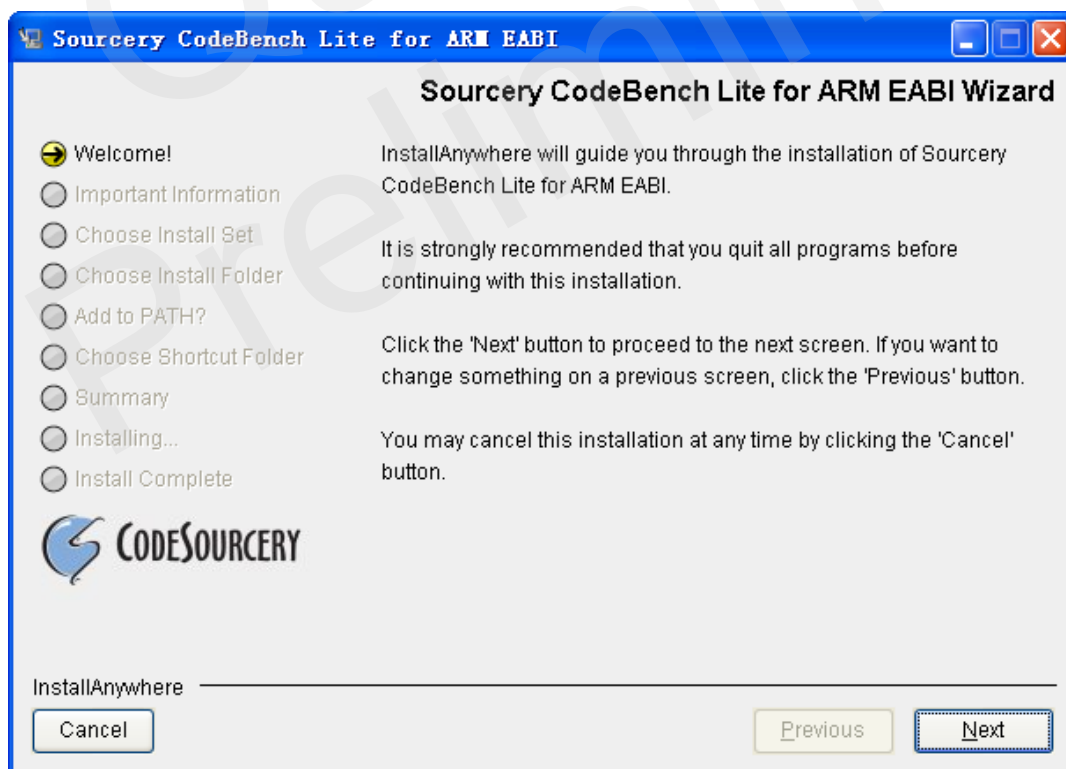


Figure 7: Running the Installer

Step 2: Choose Install Set - Select the typical install set.

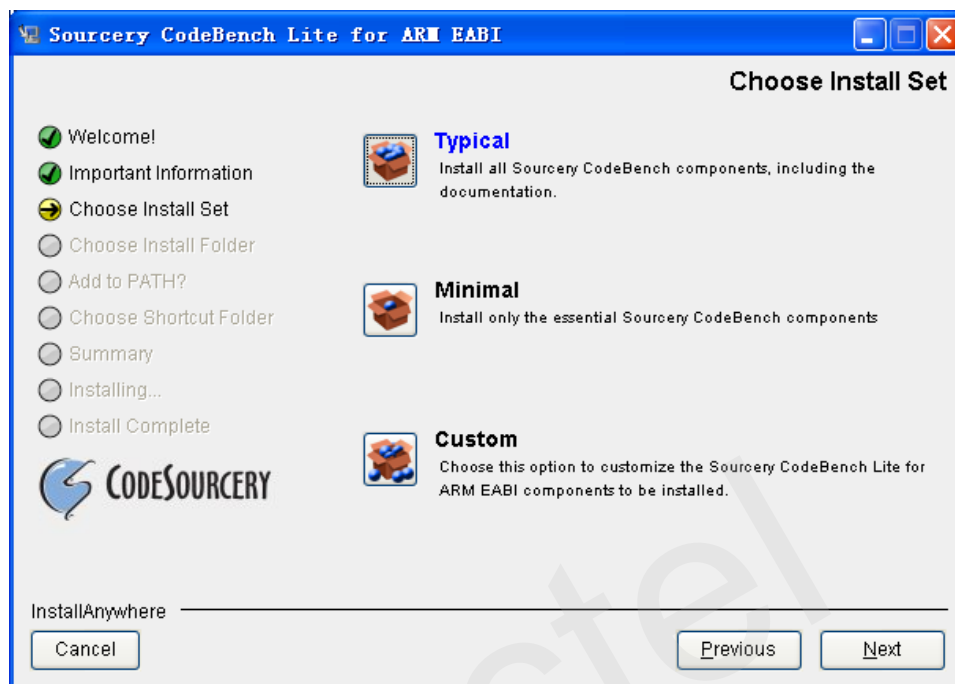


Figure 8: Choose Install Set

Step 3: Choose Install Folder - You may want to change the install directory pathname and customize the shortcut installation.



Figure 9: Choose Install Folder

Step 4: Add Product to the PATH - Keep the default choice to allow the installer to set environment variable.

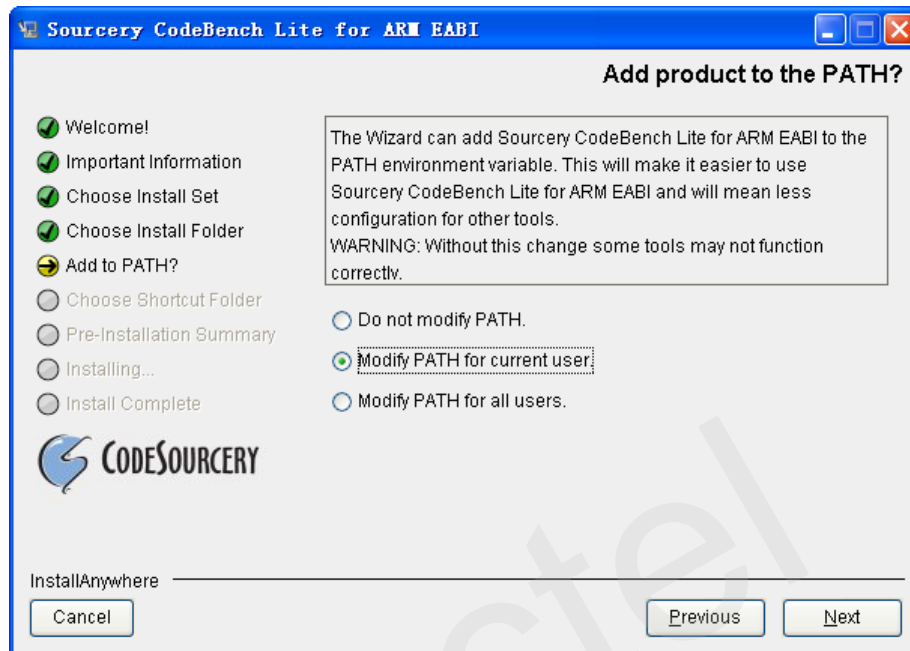


Figure 10: Add Product to the PATH

Step 5: Choose Shortcut Folder - You can customize where the installer creates shortcuts for quick access to Sourcery CodeBench Lite. When the installer has finished, it asks if you want to launch a viewer for the Getting Started guide. Finally, the installer displays a summary screen to confirm a successful install before it exits.



Figure 11: Choose Shortcut Folder

Step 6: Click “Next” to Complete Installation.

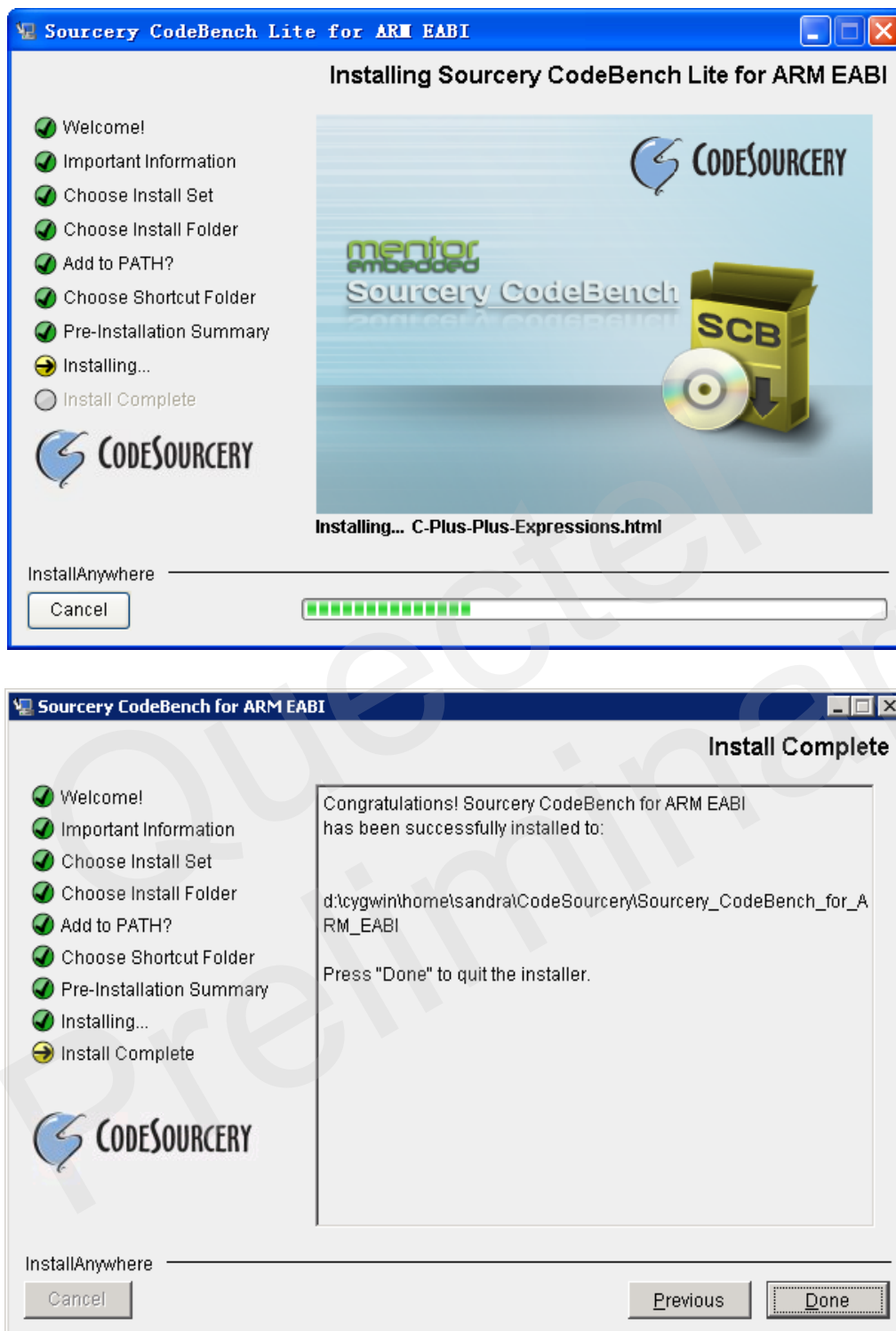


Figure 12: Complete Installation

2.4. Verify Installation

By default, the GCC installer sets up the environment variable. You can verify that your PATH is set up correctly by starting a new cmd.exe shell and running:

```
arm-none-eabi-gcc -v
```

Verify that the last line of the output contains “**Sourcery CodeBench Lite 2012.09-63**”.



```
x86-mingw32/arm-none-eabi/bin
Thread model: single
gcc version 4.7.2 (Sourcery CodeBench Lite 2012.09-63)
C:\>
```

Figure 13: Sourcery CodeBench Lite 2012.09-63

If not, you can manually set up the environment using the following command in a **cmd.exe** shell:

```
SET PATH=%PATH%; "install\bin"
```

For example:

```
SET PATH=%PATH%; "D:\Program Files\CodeSourcery\Sourcery_CodeBench_Lite_for_ARM_EABI\bin"
```

And then verify again:

```
arm-none-eabi-gcc -v
```

3 Configuration

Before compiling App with Sourcery CodeBench Lite, you have to configure the installation path and the GCC environment library path in OpenCPU SDK.

3.1. Configure the Installation Path

Open `SDK\make\gcc\gcc_makefile`, and change the value of `GCC_INSTALL_PATH` accordingly.

```
#-----  
# Configure GCC installation path, and GCC version.  
# To execute "arm-none-eabi-gcc -v" in command line can get the current gcc version  
#-----  
GCC_INSTALL_PATH=D:/Program Files/CodeSourcery/Sourcery_CodeBench_Lite_for_ARM_EABI  
GCC_VERSION=4.7.2
```

Figure 14: Change the Value of GCC_INSTALL_PATH

3.2. Configure GCC version

If you don't install the version "arm-2012.09-63-arm-none-eabi" that builds in GCC version 4.7.2 (use "arm-none-eabi-gcc -v" to check), you need to change the value of `GCC_VERSION` accordingly in the file `SDK\make\gcc\gcc_makefile`.

```
#-----  
# Configure GCC installation path, and GCC version.  
# To execute "arm-none-eabi-gcc -v" in command line can get the current gcc version  
#-----  
GCC_INSTALL_PATH=D:/Program Files/CodeSourcery/Sourcery_CodeBench_Lite_for_ARM_EABI  
GCC_VERSION=4.7.2
```

Figure 15: Change the Value of GCC_VERSION

Here, “4.7.2” indicates the GCC version, which should be corresponding to the version of the currently installed GCC (you can check the current GCC version using “arm-none-eabi-gcc -v” in command line). Then the compiler can search the correct path during compiling.

```
86-mingw32/arm-none-eabi/bin  
Thread model: single  
gcc version 4.7.2 (Sourcery CodeBench Lite 2012.09-63)  
C:\>
```

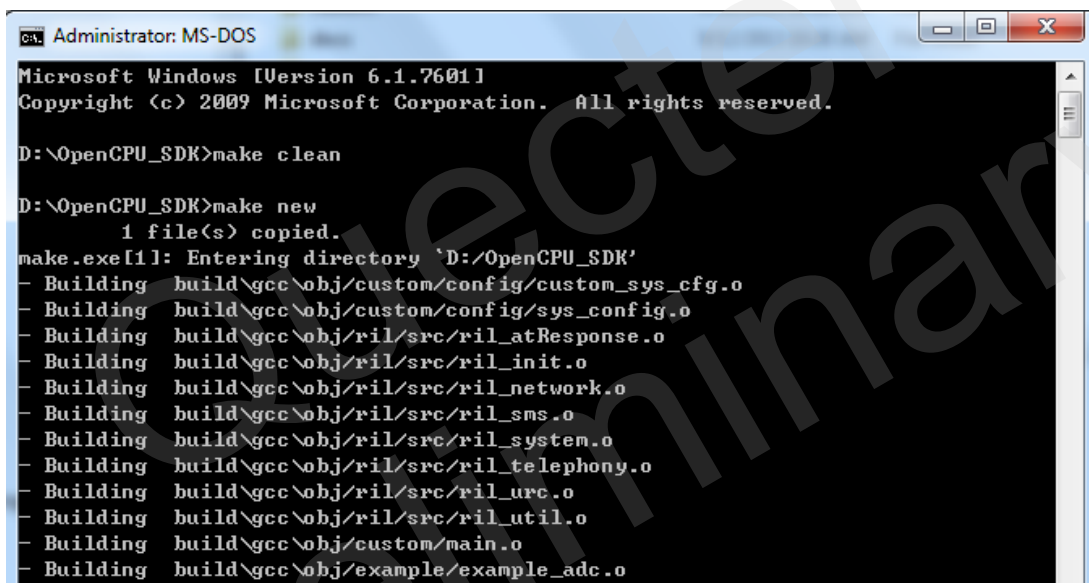
Figure 16: GCC Version

4 Compile

Now, the GCC compiling environment is set up successfully. In OpenCPU, compiling commands are executed in command line. The compiling and clean commands are defined as below.

```
make clean  
make new
```

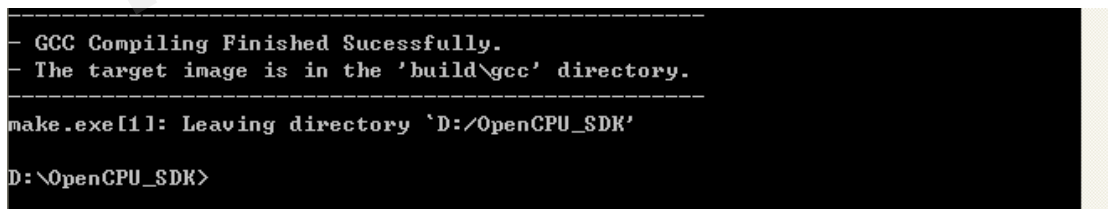
The compiling and clean commands need to be executed in the root directory of SDK.



```
Administrator: MS-DOS  
Microsoft Windows [Version 6.1.7601]  
Copyright (c) 2009 Microsoft Corporation. All rights reserved.  
  
D:\OpenCPU_SDK>make clean  
  
D:\OpenCPU_SDK>make new  
1 file(s) copied.  
make.exe[1]: Entering directory 'D:\OpenCPU_SDK'  
- Building build\gcc\obj\custom\config\custom_sys_cfg.o  
- Building build\gcc\obj\custom\config\sys_config.o  
- Building build\gcc\obj\ril\src\ril_atResponse.o  
- Building build\gcc\obj\ril\src\ril_init.o  
- Building build\gcc\obj\ril\src\ril_network.o  
- Building build\gcc\obj\ril\src\ril_sms.o  
- Building build\gcc\obj\ril\src\ril_system.o  
- Building build\gcc\obj\ril\src\ril_telephony.o  
- Building build\gcc\obj\ril\src\ril_urc.o  
- Building build\gcc\obj\ril\src\ril_util.o  
- Building build\gcc\obj\custom\main.o  
- Building build\gcc\obj\example\example_adc.o
```

Figure 17: Execute Compiling and Clean Commands

After successfully compiling, you will watch the output shown as below.



```
- GCC Compiling Finished Sucessfully.  
- The target image is in the 'build\gcc' directory.  
-----  
make.exe[1]: Leaving directory 'D:\OpenCPU_SDK'  
  
D:\OpenCPU_SDK>
```

Figure 18: Output of Successful Compiling

Please refer to “OpenCPU User Guide” document for more information about development environment.